

Binary Image Classification with CBAM-Enhanced ResNet, GeM Pooling and Edge-Map Preprocessing

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CS5480 Deep Learning Hackathon 2026

Problem Statement

Competition

IITH DL Hackathon 2026 (CS5480)

Binary image classification — **no pretrained weights**
18,000 training images, trained entirely from scratch

Classification Rule

- **Class 1:** scene has ≥ 1 sphere **AND** ≥ 1 cube
- **Class 0:** all other configurations

The signal is **purely shape** — colour & background are identical across classes

Two key challenges

- 1 **Destructive augmentation**
RandomResizedCrop can exclude the only sphere or cube → silent label corruption
- 2 **Colour bias**
Same colours appear in both classes; RGB wastes capacity on spurious cues

Dataset

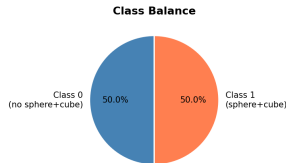
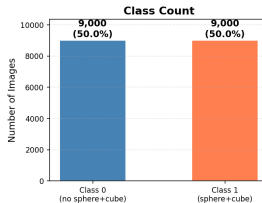
Structure

- **18,000** labelled training images (9k Class-0 + 9k Class-1, balanced)
- **5,010** unlabelled test images
- CLEVR-style 3-D renders: 4 geometric objects on a plain grey background

Train / Val split

85% / 15% random permutation
`torch.randperm, split_seed=42`

Training Dataset Distribution



ShapePreprocess: Edge-Map Preprocessing

Pipeline (deterministic, every image)

RGB $\rightarrow$ Grayscale $\rightarrow$ $\times 3$ Contrast $\rightarrow$
FIND_EDGES $\rightarrow$ RGB

Why it works

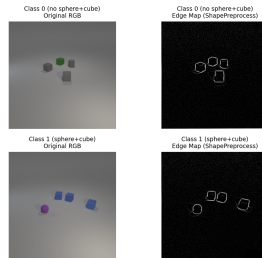
- **Sphere** $\rightarrow$ smooth circular contour
- **Cube** $\rightarrow$ sharp rectangular corners
- **Cylinder** $\rightarrow$ rounded top, flat sides

Eliminates colour & shading entirely.

Normalisation: $\mu = \sigma = [0.5, 0.5, 0.5]$

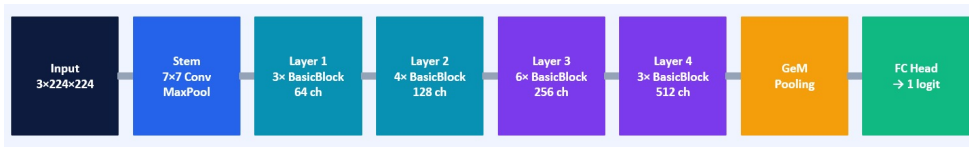
Augmentations (train only)

ShapePreprocess: RGB Input vs. Edge-Map Output



Left: original RGB. *Right:* edge map. Class 1 (bottom) shows circular sphere outline + rectangular cube corners.

Architecture Overview



Stem (Conv 7×7 -BN-ReLU-MaxPool) → **4 Stages** (3-4-6-3 blocks, 64/128/256/512 ch) →
GeM Pool → **FC Head** → 1 logit

Architecture Details

BasicBlock (ResNet-34 schedule)

- 1 Conv1: 3×3 Conv-BN-ReLU
- 2 Conv2: 3×3 Conv-BN (no ReLU)
- 3 **CBAM** attention
- 4 **DropPath** (stochastic depth)
- 5 Add shortcut $\rightarrow$ **ReLU**

CBAM

Channel: $\mathbf{w}_c = \sigma(\text{MLP}(\bar{\mathbf{x}}_{\text{avg}}) + \text{MLP}(\bar{\mathbf{x}}_{\text{max}}))$

Spatial: $\mathbf{w}_s = \sigma(\text{Conv}_{7 \times 7}([\mathbf{x}_{\text{avg}}; \mathbf{x}_{\text{max}}]))$

GeM Pooling (learnable p , init 3)

$$\mathbf{f} = \left(\frac{1}{HW} \sum_{h,w} x_{h,w}^p \right)^{1/p}$$

Emphasises most-activated positions.

Outperforms ConcatPool2d on this dataset.

Stochastic Depth (DropPath)

$$p_l = \frac{l-1}{L-1} \cdot p_{\max}, \quad l = 1, \dots, 16, \quad p_{\max} = 0.1$$

First block never dropped; linearly scales to $p_{16} = 0.1$.

Classifier Head

Drop(0.2) $\rightarrow$ Linear(512,256) $\rightarrow$ ReLU $\rightarrow$ Drop(0.1) $\rightarrow$ Linear(256,1)

Training Strategy

Two-Phase Training

Phase 1 — Validation training

- Train on 85% split, early stopping (patience 20)
- Grid-search threshold on 15% held-out set (401 steps, [0.05, 0.95])

Phase 2 — Full retrain

- Same seed, 100 epochs on all 18k images
- Apply Phase-1 threshold to test predictions

Optimiser & Schedule

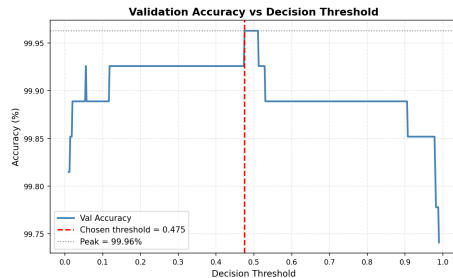
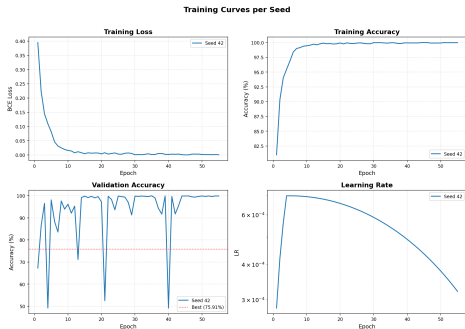
AdamW, $\eta_0 = 7 \times 10^{-4}$, $\lambda = 10^{-4}$

5 warm-up epochs + cosine annealing; grad clip 1.0; AMP

Key hyperparameters

Parameter	Value
Image size	224×224
Batch size	128
Max epochs	100
Learning rate	7×10^{-4}
Drop path rate	$0 \rightarrow 0.1$ (linear)
Dropout	0.2 / 0.1
Random seed	42 (single)

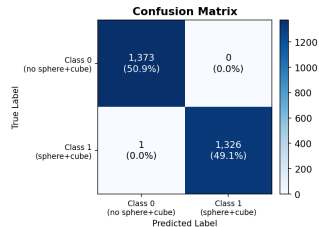
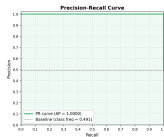
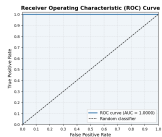
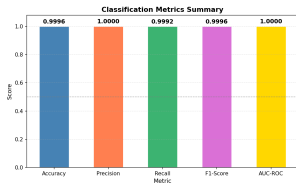
Training Dynamics & Threshold Calibration



Steady convergence; val tracks train after warm-up. Red dotted line = prior single-model best (75.91%).

Optimal threshold $\ll 0.5$; post-hoc calibration is essential for edge-map inputs.

Evaluation Metrics



Public Leaderboard Score

0.775311 (77.53%)

5,010 test predictions · seed 42

Score Progression

Key Change	Description	LB Score
Baseline	Initial submission	0.750872
GeM + Stochastic Depth	Better pooling + reg.	0.759102
Heavy Reg (MixUp, CutMix)	Over-regularised	0.752867
Rollback + TTA	Light aug, hflip TTA	0.755860
100 epochs, 3-seed ensemble	Budget + ensemble	0.768827
ShapePreprocess	Edge-map, single seed	0.775311

Key insight

Prior best 0.768827 used a **3-seed ensemble**. Final 0.775311 beats it with a **single seed** via ShapePreprocess: **better input representation > ensemble diversity**.

Key Observations

What hurts

- **RandomResizedCrop**: silently corrupts labels by excluding objects
- **MixUp / CutMix**: destroy the sharp edge signal
- **Label smoothing**: penalises confidence on an unambiguous task
- **EMA**: catastrophic failure (0.495 score)
- **3-seed ensemble** on same split: averaging adds noise on correlated models

What works

- **ShapePreprocess**: eliminates colour bias, directly exposes shape signal
- **GeM pooling**: better than average or concat pooling
- **TTA** (hflip): consistent $\approx 0.5\%$ gain
- **Threshold calibration**: optimal threshold well below 0.5 on edge-map inputs
- **Light regularisation**: dropout 0.2, drop_path 0.1

Conclusion

Summary

- Custom **ResNet-34-like** CNN (3-4-6-3 blocks) + CBAM + GeM Pooling, trained **from scratch**
- **ShapePreprocess**: highest-impact change — converts RGB to grayscale edge map, directly encoding the sphere/cube discrimination signal
- **Two-phase training**: threshold calibrated on held-out data; final model trained on all 18k images
- **Public score**: **0.775311 (77.53%)** — +2.4% over initial baseline, +0.9% over prior 3-seed ensemble

Member Contributions

Lochan: architecture (CBAM, GeM, DropPath), training pipeline, ShapePreprocess, tuning, report

Harsha: dataset analysis, classification rule discovery, augmentation experiments, validation, report

Acknowledgements

Prof. Srijith P.K. and Prof. Konda Reddy Mopuri
CS5480 DL Hackathon, IIT Hyderabad, 2026

Reproducibility

`submission.py, seed=42`

Binary Classification — CS5480